Teacher answers and topic guidance

Accept equivalent science in the usual response mode.

Starter

1. Evidence can test the possible benefit and check for harm.
2. Chromosomes. Each new nucleus receives a matching set.
3. In the nucleus.

Zoom

A chromosome contains a long DNA molecule packaged with proteins. The bracket marks a region of DNA, a gene. A represents a single base, adenine.

Fresh check

Move the gene label to the bracketed DNA section and label the letter A as a base. Many bases make up the sequence in a gene.

Arrival Supported

1. Chromosome.
2. A gene.
3. A.
4. No. A gene contains a sequence of bases.

Arrival Independent

1. Base.
2. The chromosome.
3. The section that represents a gene.
4. A base is one unit in the DNA sequence of the gene.

Arrival Aspire

1. The middle drawing.
2. The gene is one section of the DNA in a chromosome.
3. It represents one DNA base, adenine.
4. Any valid limit, such as size, the full base sequence or the DNA packaging proteins.

Arrival Choice

1. Chromosome.
2. Gene.
3. A.
4. Chromosome → gene → base.

Exit Supported

1. Gene.
2. A, C, G or T.

Exit Independent

1. A gene is a section of the DNA packaged in the chromosome.
2. It is one unit in the gene’s DNA sequence.

Exit Aspire

1. A gene is a section of DNA. A chromosome contains a long packaged DNA molecule.
2. It does not show true scale, all the bases or the packaging proteins.

Exit Choice

1. Chromosome.
2. Gene.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Label the three zoom levels: chromosome, gene and base.

Chromosome, gene, base.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete: A gene is a section of ___.

DNA.

Maximum 1 marks. Accept equivalent valid wording or a clear model demonstration.

3. Point to one base in the model and explain how it fits in the gene.

A is one base in the gene’s DNA sequence.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

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Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Explain what the gene bracket marks in the DNA drawing.

It marks a section of DNA. The gene has a sequence of bases.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why the chromosome drawing shows a larger structure than the gene drawing.

A chromosome contains a long DNA molecule packaged with proteins. A gene is one section of the DNA.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Correct “A base is a small chromosome”. Explain why.

A base is a unit of DNA. A gene has many bases, and a chromosome contains packaged DNA.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

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Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Explain what is lost if a model labels an entire chromosome as a gene.

The label loses the distinction between packaged DNA and one region. A chromosome contains many genes and other DNA.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how the base sequence relates to a protein instruction in a protein-coding gene.

The gene is a DNA section whose base sequence provides instructions for a protein.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. State one limit of the zoom model and make a justified improvement.

For example, the sketch leaves out proteins. Add a label to show that the long DNA molecule is packaged with proteins.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Which is packaged DNA: chromosome or base? Give a reason.

Chromosome. It contains a long DNA molecule packaged with proteins.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Which marks a section of DNA: gene or genome? Give a reason.

Gene. The genome is the full DNA set.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Which shows one base: A or the whole gene? Give a reason.

A. A gene contains a sequence of many bases.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt